

Practice Quiz: Perception

Part A: Multiple Choice

1. According to Active Inference, perception is best understood as: A) Passive reception of sensory data from the environment B) Unconscious inference — the brain's best guess about the causes of sensory stimulation C) Direct contact with the structure of the environment D) A purely social construction mediated by language and culture
2. Prediction error in the Active Inference framework corresponds to: A) The discrepancy between expected and actual sensory observations B) A computational bug that the brain should eliminate C) The subjective feeling of confusion D) The Bayesian prior probability of an event
3. Merleau-Ponty's concept of tool transparency is best explained in Active Inference terms as: A) The tool becoming invisible to the visual system B) The absence of prediction error during skilled, fluent tool use C) A repression of tool-awareness from consciousness D) The tool being incorporated into the agent's physical boundary
4. Hoffman's user-interface theory of perception claims that: A) Perception accurately reproduces the structure of objective reality B) Perception provides species-specific icons optimized for fitness, not truth C) Only humans have genuine perceptual experiences D) Virtual reality is indistinguishable from real perception
5. The "dark room problem" challenges Active Inference by asking: A) Why organisms don't seek out complete darkness to minimize surprise B) How agents perceive in low-light conditions C) Whether Markov Blankets exist in dark environments D) If unconscious perception is possible during sleep
6. The philosophical distinction between perception and hallucination, under Active Inference, is: A) Absolute — perception involves real objects, hallucination does not B) A matter of degree — hallucinations are predictions without sufficient sensory constraint C) Meaningless — the framework cannot distinguish them D) Based solely on the agent's verbal reports

7. Gibson's ecological psychology differs from Active Inference primarily in that: A) Gibson denied the existence of information in the environment B) Gibson argued perception requires internal generative models C) Gibson claimed perception is direct — based on invariant information structures, not inference D) Gibson rejected the role of action in perception

Part B: Short Answer

1. Explain the relationship between prediction error and the phenomenological concept of "breakdown" (Heidegger). Give an example from everyday experience.
2. Can an Active Inference agent perceive something that is not there (a hallucination) or fail to perceive something that is there (inattentional blindness)? How does the framework explain each?
3. Compare the Kantian thesis that "the mind constitutes the experienced world" with the Active Inference claim that "perception is shaped by the generative model." Are they making the same claim?